Every card you've posted, what we did with it, and what we still owe you.

Four questions across two threads. For each one: the data actually available, the method applied to it, what came out, and the next step. Working through your pump-off cards turned up a defect in our own fillage calculation — it is documented at the end, along with the fix and what it now reads instead.

CASE	INPUT DATA	METHODOLOGY	DIAGNOSIS	WAY FORWARD
Undulation on the top end of the card "DYNAMOMETER DISCUSSIONS" · JUL 2026 *"Why is there so much undulation on the top end of the card? Break it down for me!"*	**41**" stroke at surface **4,300 ft** pump depth **6.4** SPM **1¼**" top hold-down insert pump **4,200 ft** of ¾" steel rods **C-66** prime mover 23 BOPD, no water 25 psi casing · Baird valve 150 psi Card supplied as an image, not a data file	• Card digitized from the posted image, with a stated error budget • Everitt–Jennings surface-to-downhole conversion (SPE 18189) • Fillage from plunger stroke, not surface stroke	SUPERSEDED — WE GOT TWO THINGS WRONG • We quoted a gearbox rating derived from "C-66". You called it the *prime mover*. It is an engine designation and carries no gearbox rating — that was ours to get right and we didn't. • Fillage was computed off surface stroke instead of plunger stroke. • Walter Phillips also corrected section 3b on the peaks.	• Re-issue corrected, against the raw file rather than a digitized image • Still blocked on a real gearbox rating before any torque or counterbalance work • Undulation candidates to test explicitly: rod-string harmonics vs. a pound echo
Pump-off progression across five minutes "PUMP CARDS — WHAT HAPPENED?!" · AUG 2026 *"What happened to my pump card over the course of this 5 minutes?! And what things might I do to fix it?"*	Card sequence, 5-minute window, downhole view Position **0–38 in**, load **−600 to +1,800 lb** "**Not on a controller yet**" Clutch kick-out prepared by Amplified, not yet installed Screen photographs, not an export	• Pump card forward-modelled *up* the rod string to a surface card, then solved back down — so the solver performs a real conversion instead of stripping a rod string that was never there • Fillage recovered per stroke • Transfer point swept full → pumped off to trace the progression	REPRODUCED END TO END Full barrel to near-empty across the five minutes, matching what Andrew Aday read off it. Fillage now tracks monotonically down the whole pounding range, 88% to 12%, with no reversal. Getting there exposed a defect in our own corner detection — panel below — which is fixed.	• Send the raw file and we return your fillage stroke by stroke • Send one *full-pump* card too — a full pump reads 88% on this scale, not 100%, so a "shut down at 75%" rule means something different here than it sounds. That calibration has to come first. • Then a setpoint, with the barrels it costs stated separately

CASE	INPUT DATA	METHODOLOGY	DIAGNOSIS	WAY FORWARD
Noise in the bottom-left corner "PUMP CARDS — WHAT HAPPENED?!" · AUG 2026 > *"Could the noise in this corner be from gas in the pump decompressing…? There isn't really any noise here when the pump is full!"*	Derived downhole card, lower-left segment **Unknown: is the dyno accelerometer-based?** Walter asked this in-thread and it went unanswered — it decides the whole question	• Separate numerical artefact from physical signal before interpreting • Position is double-integrated from acceleration, so error accumulates within a stroke • The lower-left sits immediately after the stroke-to-stroke discontinuity, before the wave equation settles	NOT ANSWERABLE FROM THIS DATA • Walter Phillips: don't over-read that corner; reprocess first. Gas expansion is visible at the **top** left, not the bottom left. • Andrew Aday: some gas interference in the slope near fluid/plunger contact, possibly a tag present only when the pump isn't full. • Our position: the corner is where the derived card is least trustworthy, so it is the worst place to hang a diagnosis.	• Confirm the dyno type — one answer, and it settles this • We produce a trust-band card: which segments carry enough numerical confidence to diagnose on, marked directly on the card
Repeated tubing splits — and whether they're the same problem "TUBING WEAR — ROD PUMP" > *"Regular holes in tubing, typically splits. Never on the same joint. No scale or corrosion issues… does J55 wear faster than L or N80?"*	Repeated splits, **never the same joint** No scale, no corrosion Options under consideration: tubing rotators, lined tubing, better rod guides, optimising pump-off **Missing: depths of the failures**	• Rod-tubing side load from card-derived loads • Buckling / neutral-point check on the lower rods through the downstroke • Correlate split depths against the computed neutral point	HYPOTHESIS — STATED AS ONE, NOT A FINDING Fluid pound drives the lower rods into compression on the downstroke; compression buckles them; buckled rods bear on tubing. "Never the same joint" fits a neutral point that *moves* with fillage, rather than a fixed defect. If that holds, **your tubing wear and your pump-off are one problem**, and the pump-off fix is also the tubing fix. On J55 vs L80/N80: harder grades resist abrasive wear somewhat better, and J55 is the softest of the three — so directionally yes. But grade is second-order here. Side load is what's eating the tubing, and no grade survives a buckled rod indefinitely.	• Send the depths of the last several splits — that one list tests the hypothesis cheaply • If it holds: fix pump-off first, then rod guides. Rotators and lined tubing treat the symptom. • If it fails: we say so, and look at side load from wellbore deviation instead

A defect your cards found in our tooling — and the fix

Preparing this review, we swept a pump-off progression through our solver. Fillage fell cleanly as the barrel emptied and then, past a point, jumped to **99.6%** and stayed pinned there no matter how badly the pump was pounding. A pump that fills less cannot report filling more.

PUMP CONDITION	WAS	NOW
full pump (anchor)	88.1%	88.1%
pounding, mild	64.2%	64.2%
pounding, moderate	38.9%	38.9%
pounding, severe	18.8%	18.8%
pounding, worse still	**99.6%**	16.4%
pounding, worst	**99.6%**	11.9%

Cause. The bottom-right corner of the card marks where the travelling valve opens, and it sets net stroke. Our detector was picking a corner at the *top* of the downstroke, where no fluid load had transferred at all — so net stroke collapsed onto gross stroke and the arithmetic returned "full".

Fix. A bottom-right corner is now rejected unless at least 35% of the downstroke's load drop has actually occurred there. On the four vendor-analysed reference wells the correct corner realises 0.675 to 0.866, and the failure realises 0.000 — so the guard cannot fire on a card we have evidence for. Those four wells still agree with the vendor analyses to within 0.85% fillage, unchanged.

Why it mattered to you specifically. You wrote that these wells "don't pump off slowly — when they go for it, they go." Severe, fast pump-off is precisely the regime where this failed, and it failed *toward healthy*: a controller setpoint driven off that number would have sat reading 99.6% while the pump slammed. It is the reason we would not quote you a setpoint last week, and the reason we can now.

Found and fixed 5 Aug 2026. It had survived our existing tests because the assertions in place only checked fillage was between 0 and 100, which 99.6% satisfies. There is now a regression test that sweeps the pump from full to pounded and fails if the reported fillage ever rises.

Also noted: the tally app

You offered $500 for an iPhone app that scans a handwritten tally sheet to Excel — camera capture, works offline, manual correction for illegible digits. That's a well-specified brief, and offline handwriting capture on a phone is now a solved problem.

We'd rather build it than bid on it. If it's still open, say so and we'll put a working build in front of you — you can decide afterwards whether it was worth anything.

WHAT WE NEED FROM YOU

- **Raw card files** from both threads — any export format beats a photo of the screen.
- **One full-pump card** from the same well, so the fillage scale can be calibrated before any threshold is set.
- **The gearbox make, model and rating.** The C-66 is the prime mover, so we still have nothing to check torque or counterbalance against.
- **Is the pump-off well the same well** as the 41" / 4,300 ft / 6.4 SPM one? Both mention a C-66, but that doesn't make them the same well and we're not going to assume it.
- **Rod string is 4,200 ft, pump is at 4,300 ft** — what's the missing 100 ft?
- **Is the dyno accelerometer-based?**
- **Depths of the last several tubing splits.**

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Quotes are from the operator's own public posts. Solver: Everitt–Jennings space-marching finite difference (SPE 18189). Figures in the defect panel are from a reconstruction using the stated well parameters — they demonstrate the failure mode, they are not measurements of this well.